KAIZEN MATHS

Lesson starter

Roots of Equations and Transformations of Roots

Cubic Roots

Name _____

S1

Use the cubic relationships between coefficients and roots.

The roots of $x^3 - 4x^2 + 5x - 2 = 0$ are α, β and γ. Find $\alpha + \beta + \gamma$, $\alpha\beta + \alpha\gamma + \beta\gamma$, and $\alpha\beta\gamma$.

S2

Form each cubic equation from the given roots.

Form a cubic equation with roots 1, -5 and -1.

S3

Use the given root information to find the unknown cubic coefficients.

The equation $x^3 + px^2 + qx + r = 0$ has roots α, β and γ with $\alpha + \beta + \gamma = 12$, $\alpha\beta + \alpha\gamma + \beta\gamma = 47$, and $\alpha\beta\gamma = 60$. Find p, q and r.

S4

Use $\alpha + \beta + \gamma$ and $\alpha\beta + \alpha\gamma + \beta\gamma$ to evaluate each expression.

The roots of $x^3 - 10x^2 + 31x - 30 = 0$ are α, β and γ. Find $\alpha^2 + \beta^2 + \gamma^2$.

Show your working

KAIZEN MATHS

Lesson starter

Roots of Equations and Transformations of Roots

Cubic Roots

Name _____

S1

Use the cubic relationships between coefficients and roots.

The roots of $x^3 - 4x^2 + 5x - 2 = 0$ are α, β and γ. Find $\alpha + \beta + \gamma$, $\alpha\beta + \alpha\gamma + \beta\gamma$, and $\alpha\beta\gamma$.

S2

Form each cubic equation from the given roots.

Form a cubic equation with roots 1, -5 and -1.

S3

Use the given root information to find the unknown cubic coefficients.

The equation $x^3 + px^2 + qx + r = 0$ has roots α, β and γ with $\alpha + \beta + \gamma = 12$, $\alpha\beta + \alpha\gamma + \beta\gamma = 47$, and $\alpha\beta\gamma = 60$. Find p, q and r.

S4

Use $\alpha + \beta + \gamma$ and $\alpha\beta + \alpha\gamma + \beta\gamma$ to evaluate each expression.

The roots of $x^3 - 10x^2 + 31x - 30 = 0$ are α, β and γ. Find $\alpha^2 + \beta^2 + \gamma^2$.

ALG-4282-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Roots of Equations and Transformations of Roots

Cubic Roots

Name _____

EA1

Use the cubic relationships between coefficients and roots.

The roots of $x^3 - 7x^2 + 14x - 8 = 0$ are α, β and γ. Find $\alpha + \beta + \gamma$, $\alpha\beta + \alpha\gamma + \beta\gamma$, and $\alpha\beta\gamma$.

EA2

Form each cubic equation from the given roots.

Form a cubic equation with roots -3, -5 and -1.

Show your working

KAIZEN MATHS

Exit ticket

Roots of Equations and Transformations of Roots

Cubic Roots

Name _____

EB1

Use the given root information to find the unknown cubic coefficients.

The equation $x^3 + px^2 + qx + r = 0$ has roots α, β and γ with $\alpha + \beta + \gamma = 10$, $\alpha\beta + \alpha\gamma + \beta\gamma = 29$, and $\alpha\beta\gamma = 20$. Find p, q and r.

Show your working

KAIZEN MATHS

Exit ticket

Roots of Equations and Transformations of Roots

Cubic Roots

Name _____

EA1

Use the cubic relationships between coefficients and roots.

The roots of $x^3 - 7x^2 + 14x - 8 = 0$ are α, β and γ. Find $\alpha + \beta + \gamma$, $\alpha\beta + \alpha\gamma + \beta\gamma$, and $\alpha\beta\gamma$.

EA2

Form each cubic equation from the given roots.

Form a cubic equation with roots -3, -5 and -1.

ALG-4282-L2 · EA

Show your working

KAIZEN MATHS

Exit ticket

Roots of Equations and Transformations of Roots

Cubic Roots

Name _____

EB1

Use the given root information to find the unknown cubic coefficients.

The equation $x^3 + px^2 + qx + r = 0$ has roots α, β and γ with $\alpha + \beta + \gamma = 10$, $\alpha\beta + \alpha\gamma + \beta\gamma = 29$, and $\alpha\beta\gamma = 20$. Find p, q and r.

ALG-4282-L2 · EB

Show your working

Teacher answers & working

Roots of Equations and Transformations of Roots · Cubic Roots

ALG-4282-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

Use the cubic relationships between coefficients and roots.

The roots of $x^3 - 4x^2 + 5x - 2 = 0$ are α, β and γ. Find α + β + γ, αβ + αγ + βγ, and αβγ.

Answer:

$\alpha + \beta + \gamma = 4, \alpha\beta + \alpha\gamma + \beta\gamma = 5, \alpha\beta\gamma = 2$

Working:

For $ax^3 + bx^2 + cx + d = 0$:

$\alpha + \beta + \gamma = \frac{-b}{a}$, $\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$, and $\alpha\beta\gamma = \frac{-d}{a}$.

For $x^3 - 4x^2 + 5x - 2 = 0$, $a = 1$.

So $\alpha + \beta + \gamma = 4$.

$\alpha\beta + \alpha\gamma + \beta\gamma = 5$.

$\alpha\beta\gamma = 2$.

S2

Form each cubic equation from the given roots.

Form a cubic equation with roots 1, -5 and -1.

Answer:

$x^3 + 5x^2 - x - 5 = 0$

Working:

For roots α, β and γ, the monic cubic is $x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \alpha\gamma + \beta\gamma)x - \alpha\beta\gamma = 0$.

$\alpha + \beta + \gamma = 1 + -5 + -1 = -5$.

$\alpha\beta + \alpha\gamma + \beta\gamma = (1)(-5) + (1)(-1) + (-5)(-1) = -1$.

$\alpha\beta\gamma = (1)(-5)(-1) = 5$.

Therefore $x^3 + 5x^2 - x - 5 = 0$.

S3

Use the given root information to find the unknown cubic coefficients.

The equation $x^3 + px^2 + qx + r = 0$ has roots α, β and γ with α + β + γ = 12, αβ + αγ + βγ = 47, and αβγ = 60. Find p, q and r.

Answer:

$p = -12, q = 47, r = -60$

Working:

For $x^3 + px^2 + qx + r = 0$:

$\alpha + \beta + \gamma = -p, \alpha\beta + \alpha\gamma + \beta\gamma = q$, and $\alpha\beta\gamma = -r$.

$-p = 12$, so $p = -12$.

$q = 47$.

$-r = 60$, so $r = -60$.

Teacher answers & working

Roots of Equations and Transformations of Roots · Cubic Roots

ALG-4282-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S4

Use α + β + γ and αβ + αγ + βγ to evaluate each expression.

The roots of $x^3 - 10x^2 + 31x - 30 = 0$ are α, β and γ. Find $\alpha^2 + \beta^2 + \gamma^2$.

Answer:

38

Working:

Use $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$.

From the equation, α + β + γ = 10.

Also αβ + αγ + βγ = 31.

So $\alpha^2 + \beta^2 + \gamma^2 = (10)^2 - 2(31)$.

$= 100 - 62 = 38$.

EA1

Use the cubic relationships between coefficients and roots.

The roots of $x^3 - 7x^2 + 14x - 8 = 0$ are α, β and γ. Find α + β + γ, αβ + αγ + βγ, and αβγ.

Answer:

$\alpha + \beta + \gamma = 7, \alpha\beta + \alpha\gamma + \beta\gamma = 14, \alpha\beta\gamma = 8$

Working:

For $ax^3 + bx^2 + cx + d = 0$:

$\alpha + \beta + \gamma = \frac{-b}{a}$, $\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$, and $\alpha\beta\gamma = \frac{-d}{a}$.

For $x^3 - 7x^2 + 14x - 8 = 0$, $a = 1$.

So $\alpha + \beta + \gamma = 7$.

$\alpha\beta + \alpha\gamma + \beta\gamma = 14$.

$\alpha\beta\gamma = 8$.

EA2

Form each cubic equation from the given roots.

Form a cubic equation with roots -3, -5 and -1.

Answer:

$x^3 + 9x^2 + 23x + 15 = 0$

Working:

For roots α, β and γ, the monic cubic is $x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \alpha\gamma + \beta\gamma)x - \alpha\beta\gamma = 0$.

$\alpha + \beta + \gamma = -3 + -5 + -1 = -9$.

$\alpha\beta + \alpha\gamma + \beta\gamma = (-3)(-5) + (-3)(-1) + (-5)(-1) = 23$.

$\alpha\beta\gamma = (-3)(-5)(-1) = -15$.

Therefore $x^3 + 9x^2 + 23x + 15 = 0$.

Teacher answers & working

Roots of Equations and Transformations of Roots · Cubic Roots

ALG-4282-L2 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EB1

Use the given root information to find the unknown cubic coefficients.

The equation $x^3 + px^2 + qx + r = 0$ has roots α, β and γ with $\alpha + \beta + \gamma = 10$, $\alpha\beta + \alpha\gamma + \beta\gamma = 29$, and $\alpha\beta\gamma = 20$. Find p, q and r.

Answer:

$p = -10$, $q = 29$, $r = -20$

Working:

For $x^3 + px^2 + qx + r = 0$:

$\alpha + \beta + \gamma = -p$, $\alpha\beta + \alpha\gamma + \beta\gamma = q$, and $\alpha\beta\gamma = -r$.

$-p = 10$, so $p = -10$.

$q = 29$.

$-r = 20$, so $r = -20$.